

NILOY SAHA

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EDUCATION

Hofstra University

B.S. in Computer Science

New York, USA

Expected Graduation, May 2027

- o **GPA:** 3.5/4.00, *Dean's List*
- o **Related Coursework:** Data Structures & Algorithms, Computer Organization & Programming, Software Engineering, Discrete Structures and Mathematics, Object-Oriented Programming, Applied NLP and Text Mining

PROJECTS

Credit Card Fraudulent Predictor | [GitHub Link](#)

- Built a credit card fraud detection system using Python, Pandas, NumPy, and scikit-learn on 50,000+ transactions with 8 key features, preprocessing and normalizing data, and handling class imbalance with SMOTE.
- Achieved **99%+ ROC-AUC** and **0.99 F1-score** by training and tuning Random Forest and Gradient Boosting classifiers, enabling real-time fraud prediction for new credit card transactions.
- Generated visual insights, including transaction amount and class distributions using Matplotlib, to support model interpretability and feature importance analysis.

CourseCrafter AI | [GitHub Link](#) | [Site Link](#)

- Developed a full-stack Node.js & Express web app that generates structured AI learning courses from user-provided topics.
- Reduced **API response times by 30%** by integrating OpenAI API and YouTube Data API with Axios and optimizing backend performance in a Node.js & Express application.
- Delivered a modern, responsive UI using EJS templating and Bootstrap, ensuring seamless, accessible learning experiences across devices.

EXPERIENCE

Resident Success Representative

Hofstra University

New York, USA

Jan 2025 - Apr 2025

- Assisted and guided university residents by resolving housing and campus-related issues, ensuring a safe and supportive living environment.
- Coordinated events, workshops, and community programs that fostered engagement and improved resident satisfaction across dormitories.
- Collaborated with university staff to address student concerns, track incidents, and implement solutions, contributing to a well-organized and responsive residential community.

ACTIVITIES AND LEADERSHIP

HOFSTRA C.O.D.E

Member

New York, USA

Sept 2024 - Present

- Participated in weekly coding challenges and hackathons to strengthen problem-solving and collaboration skills.
- Engaged in peer-led workshops to explore emerging technologies and share practical insights with peers.
- Contributed to open-source projects and group coding initiatives, applying learned concepts to real-world scenarios.

SKILLS

Programming: Java, Python, JavaScript, HTML/CSS, SQL, Node.js, React.js, Dart, C++

Concepts: Data Structures and Algorithms, Object-Oriented Programming, Agile Methodologies, RESTful APIs, Natural Language Processing, Version Control, SDLC, Machine Learning

Tools: Android Studio, IntelliJ, PyCharm, Eclipse, AWS, Jupyter Notebooks, Git, Bootstrap, Spring Boot, Agile, VS Code, GitHub, Postman, XCode, Unreal Engine

Certification

The Complete Full-Stack Web Development Bootcamp

By Udemy

Date Issued: 20 Aug 2025

Completion Certificate: <https://www.udemy.com/certificate/UC-60a6fb8e-10a6-4e5c-8f3b-b6ede6670c6>